Statistics

- indicates numerical data arising from uncontrolled factors.
- Scientific methods which are meant for collection, analysis and interpretation of numerical data.

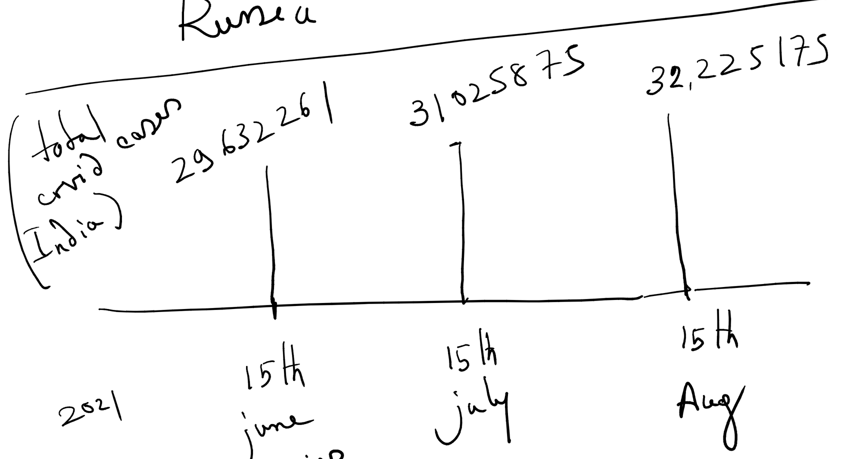
Based on presentation data can be classified as frequency data and non-frequency data.

Non-frequency data

identity of individual values has to be retained.

	total cases (covid)
USA	37 466718
India	32 225 513
Brazil	20 364 599
Russia	6 621 601

Spatial series data



time series data

frequency data

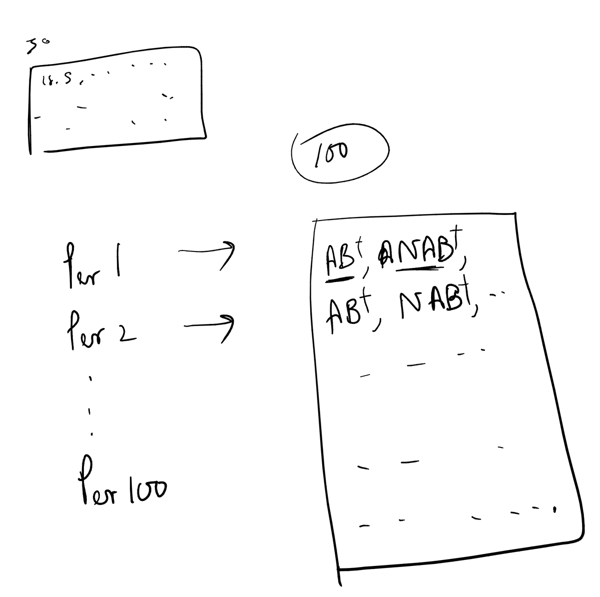
identity of individuals is unimportant and can be ignored.

- character —
  - attribute — can not be expressed numerically
  - variable — can be expressed numerically.
    - discrete — can take only some isolated values.
    - continuous — can take any value within a certain range.

freq distribution of attribute

100 patients

	freq
AB^+ blood group	25
not AB^+	75
	100



freq dist of discrete variable :

Daily number of car accident	frequency	Cumulative frequency
3	5	5
4	9	14 (less than equals to 4 accidents occur).
5	11	25
6	4	29
7	1	30
	30	

Class-1

Class-2

Frequency distribution of continuous variable:

Class interval	frequency
9.1 — 16.5	1
16.6 — 24.0	5
24.1 — 31.5	18
31.6 — 39.0	21
39.1 — 46.5	2
46.6 — 54.0	3
total	50

lower limit

upper limit

- each class contain at least 1 data point
- each of of given values is included in one of the classes
- the classes should be non-overlapping.

18.5
9.1

9.1 represents any value between 9.05 and 9.15

16.5 represents any value between 16.45 and 16.55.

(9.1 — 16.5) stands for (9.05 — 16.55).

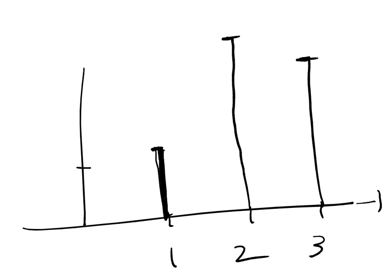
	freq
9.05 — 16.55	1
16.55 — 24.05	5
24.05 — 31.55	18
31.55 — 39.05	21
39.05 — 46.55	2
46.55 — 54.05	3
	50

upper class boundary

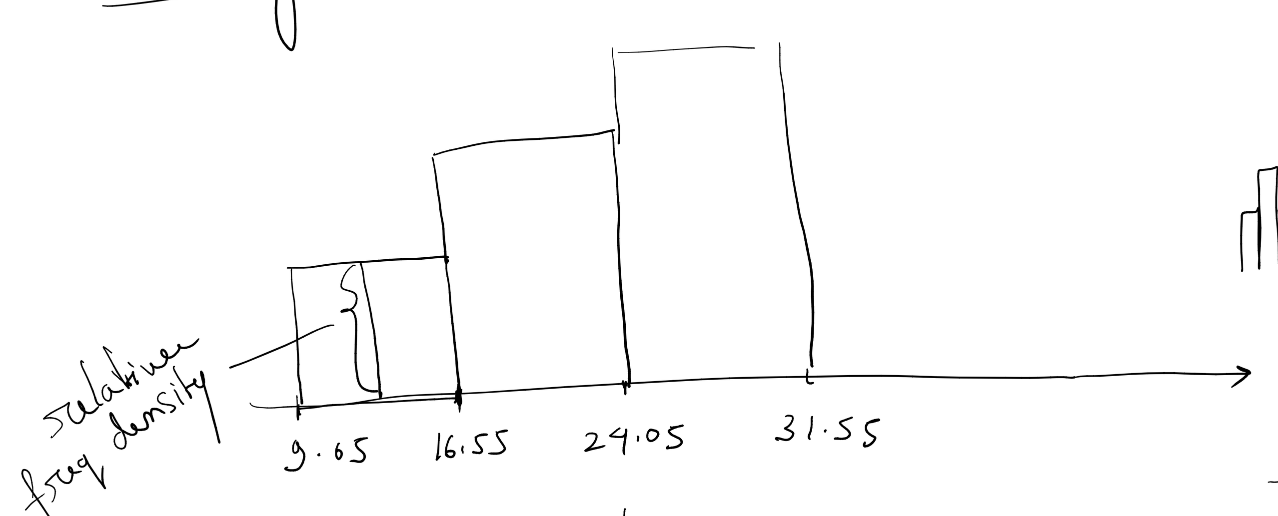
lower class boundary

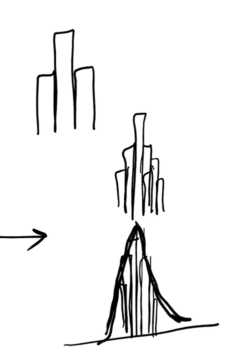
9.26 — 16.13
9.255 — 16.135
boundaries

9.12569
9.125685



Histogram:





area = 1

frequency density of a class = $\frac{\text{freq}}{\text{width of the class}}$

$\left(\frac{1}{7.5} \text{ for } 9.05 - 16.55\right)$

relative frequency density of a class = $\frac{\text{freq density of a class}}{\text{total numbers of observations}}$

$\left(\frac{1}{7.5 \times 50} \text{ for } (9.05 - 16.55)\right)$